

Computer Science & Information Technology

Computer Networks

Application Layer

DPP: 1

- Q1** Identify which name server(s) does not belong to DNS hierarchy ?
(A) Root Name Server
(B) Top Level Domain Name Server
(C) Real Name Server
(D) Authoritative Name Server
- Q2** Consider the resolution of the domain name "www.iitd.ac.in" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____ .
- Q3** Consider the resolution of the domain name "www.google.com" by a DNS resolver. Assume that iterative query mechanism is used in the resolution. Let RTT denote the round trip time between DNS resolver and DNS servers in the DNS hierarchy. The number of RTTs required in completely resolving the domain name is _____ .
- Q4** Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP. The web page contains base HTML file and four very small image objects. How many TCP connections required to load the web page completely ?
- Q5** Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP (no any parallel TCP connections). The web page contains base HTML file and four very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely?
- Q6** Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP that support parallel TCP connections. The web page contains base HTML file and four very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?
- Q7** Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP without pipeline. The web page contains base HTML file and four very small image objects. Assume base HTML file and all four image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?
- Q8** Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP with pipeline. The web page contains base HTML file and four very small image objects. Assume base HTML file and all four image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?
- Q9** Which is/are valid HTTP request method ?
(A) GET (B) PUT



(C) POST

(D) HEAD

Q10 Consider a web browser at client machine made a request for a web page to a web server.

Identify the correct sequence in which the following protocols are used by the client host?

(A) HTTP, TCP, DNS

(B) DNS, TCP, HTTP

(C) DNS, HTTP, TCP

(D) TCP, DNS, HTTP



[Android App](#) | [iOS App](#) | [PW Website](#)

Answer Key

Q1 (C)

Q2 4~4

Q3 3~3

Q4 5~5

Q5 10~10

Q6 4~4

Q7 6~6

Q8 3~3

Q9 (A, B, C, D)

Q10 (B)



[Android App](#) | [iOS App](#) | [PW Website](#)

Hints & Solutions

Q1 Text Solution:

Real Name Server

Q2 Text Solution:

Domain Name : "www.iitd.ac.in"

Mode : Iterative Query

DNS Resolver → 4-tier DNS hierarchy

Q3 Text Solution:

Domain name : "www.google.com"

Mode : Iterative Query

DNS Resolver → 3-tier DNS hierarchy

Q4 Text Solution:

Non persistent HTTP:

Web page = [Base HTML file + 4 Small image objects]

No. of TCP Connection = $(1 + 4) = 5$

Q5 Text Solution:

Non-persistent HTTP:

Web page = [Base HTML File + 4 Small image objects]

Total time = $2 \text{ RTT} + 4 \times 2 \text{ RTT} = 10 \text{ RTT}$

Q6 Text Solution:

Non-persistent HTTP:

Web page = Base HTML File + 4 Very small image objects (Using parallel TCP Conn.)

Total time = $2 \text{ RTT} + 2 \text{ RTT} = 4 \text{ RTT}$

Q7 Text Solution:

Persistent HTTP:

1 RTT → for TCP connection

1 RTT → for Base HTML File

1 RTT → first object request

...

1 RTT → fourth object request

Total = 6 RTT

Q8 Text Solution:

Persistent HTTP:

1 RTT → for TCP connection

1 RTT → for Base HTML file

1 RTT → for all four objects request in pipeline

Total = 3 RTT

Q9 Text Solution:

All four options are correct.

Q10 Text Solution:

(1) DNS:

(i) DNS Query

(ii) DNS Reply

(2) TCP:

(i) TCP Connection Request

(ii) TCP Accept Response

(3) HTTP:

(i) HTTP Request Message

(ii) HTTP Response Message



[Android App](#) | [iOS App](#) | [PW Website](#)